Bloch-Wigner sequence and linear groups over finite rings

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The Bloch-Wigner exact sequence is an important result in K-theory that connects the third homology group of $\mathrm{SL}_2(\mathbb{C})$ with the Bloch group, which encodes information about scissor congruences of polyhedra in hyperbolic geometry. These concepts provide deep insights into the interplay between algebra, geometry, and topology.

Hutchinson, Mirzaii, and Pérez have extended the Bloch-Wigner sequence to other classes of commutative rings, like local rings [1, 2]. In this work, we apply these results to study the low-dimensional homology groups of SL_2 over finite commutative rings. We also discuss other achievements in the area that have been obtained using these methods.

Keywords: Bloch group, Finite rings,

References

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